

3par Storage Concepts

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Key Concepts and Terms

TERMS

- ▶ PD (Physical Disk)
- ▶ Chunklet
- ▶ Mesh-Active Cluster
- ▶ Wide Striping
- ▶ Virtual Volume

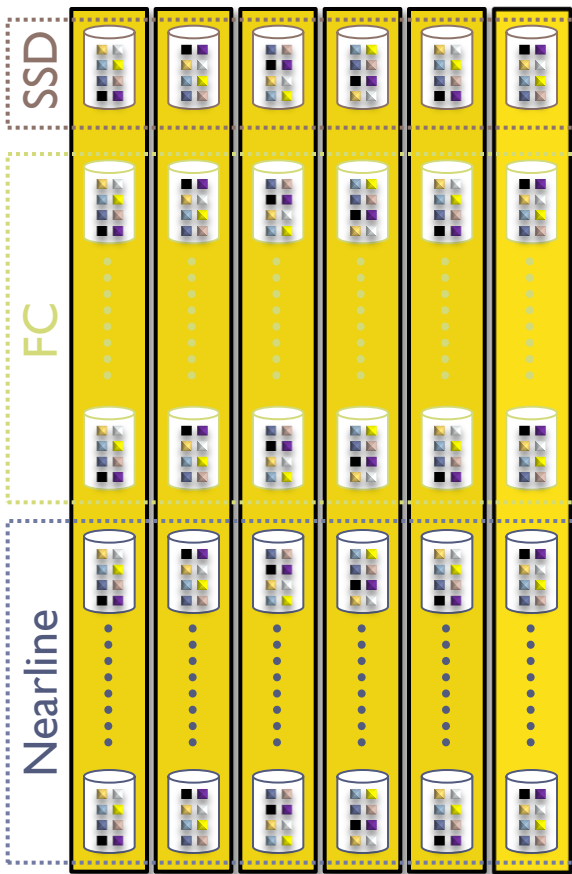
TERMS

- ▶ Virtual LUN
- ▶ CPG (Common Provisioning Group)
- ▶ Host
- ▶ VV Set, Host Set, (Autonomic Group)



HP 3PAR Virtualization – the Logical View (1 of 3)

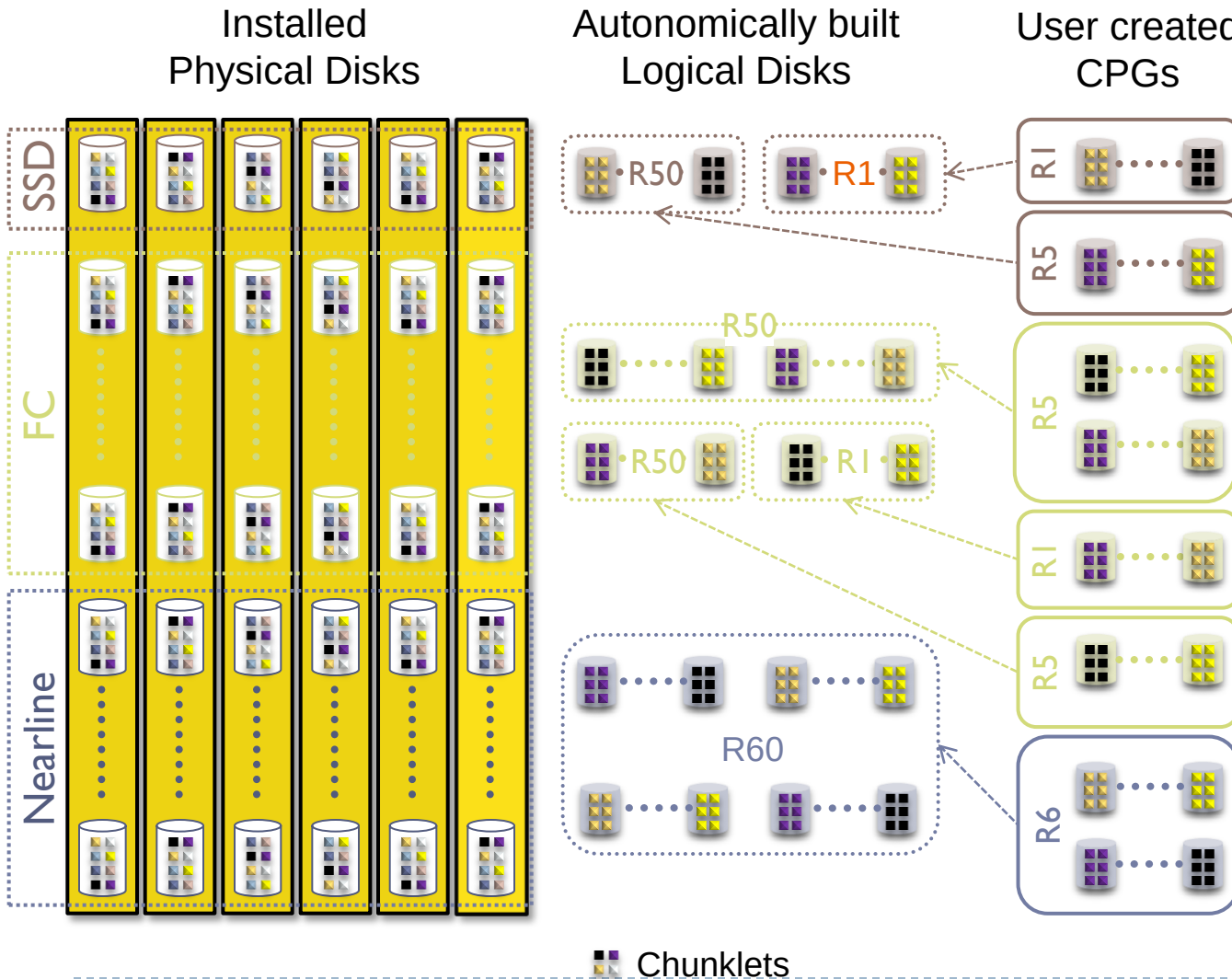
Installed
Physical Disks



 Chunklets

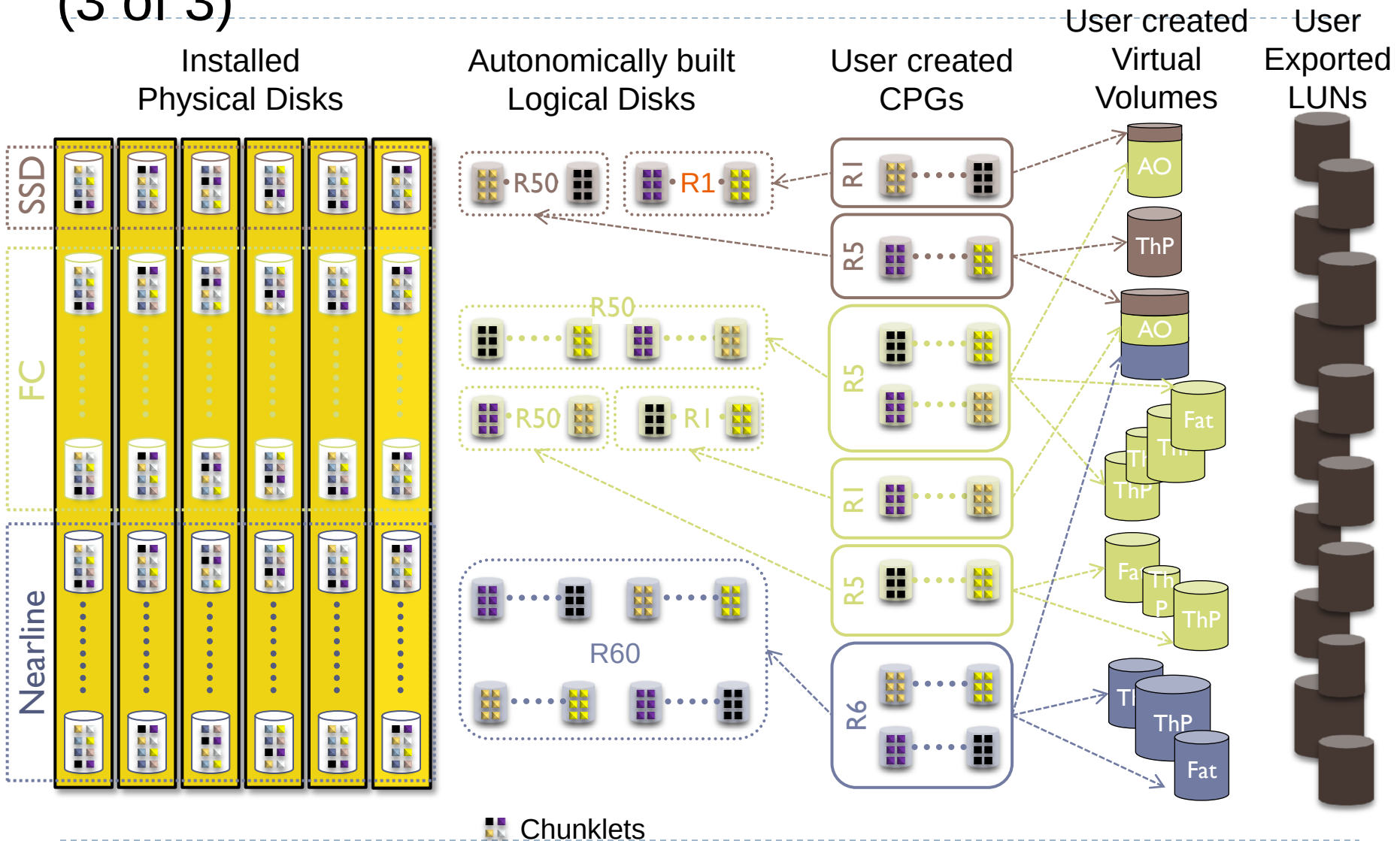
HP 3PAR Virtualization – the Logical View

(2 of 3)



HP 3PAR Virtualization – the Logical View

(3 of 3)



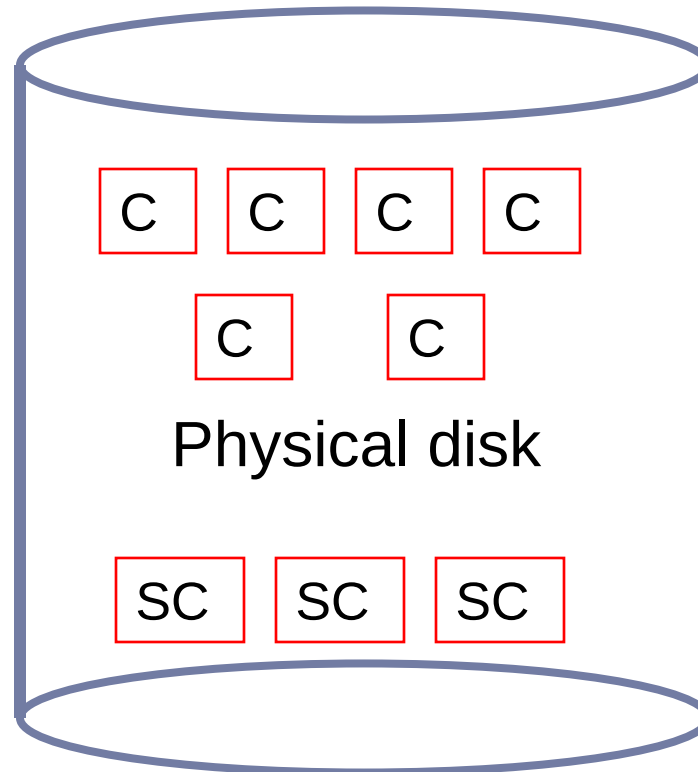
Physical Disk

Chunklets

C = Data chunklet

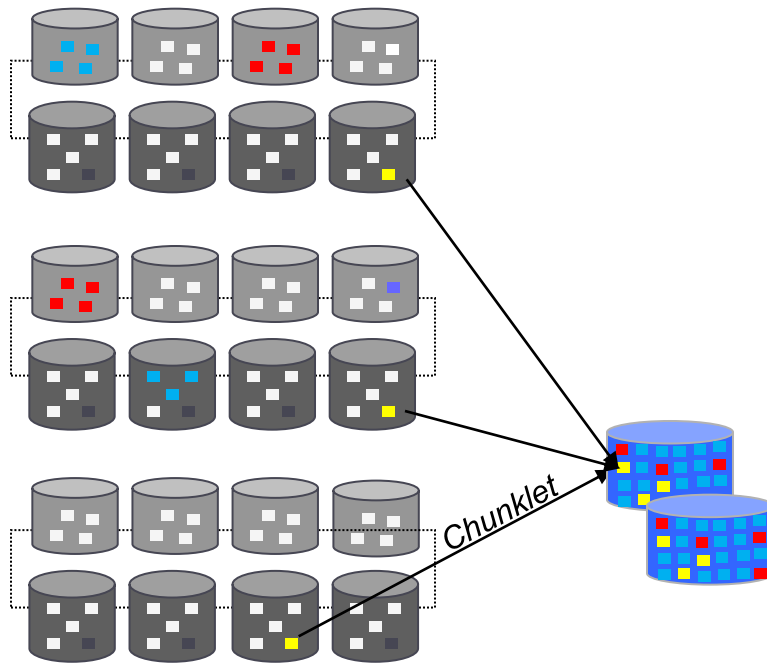
S
C = Spare chunklet

Each StorServ Physical disk is initialized with data chunklets and spare chunklets.



What is a Chunklet?

Chunklets
Physical drives broken
into fine-grained uniform
chunklets



50GB SSD, 300 and
600GB FC, 2TB NL

Device Type	Total Chunklets
50GB SSD	185
147GB FC 15K	545
300GB FC 15K	1115
400GB FC 10K	1488
450GB FC 15K	1675
500GB NL 7.2K	1861
600GB FC 15K	2234
750GB NL 7.2K	2793
1000GB NL 7.2K	3724
2000GB NL 7.2K	7225

NOTE: Total Chunklets is for an F-Class or T-Class system.

Chunklets are Automatically Grouped by Drive Rotational Speed

- ▶ SATA (Nearline) disks are usually the biggest group (depends on configuration)
- ▶ Fibre Channel disk chunklet group
- ▶ Solid State disk chunklet group is smaller and reserved for high-value I/O



Software architecture

Rapid, tailored provisioning...

Chunklets

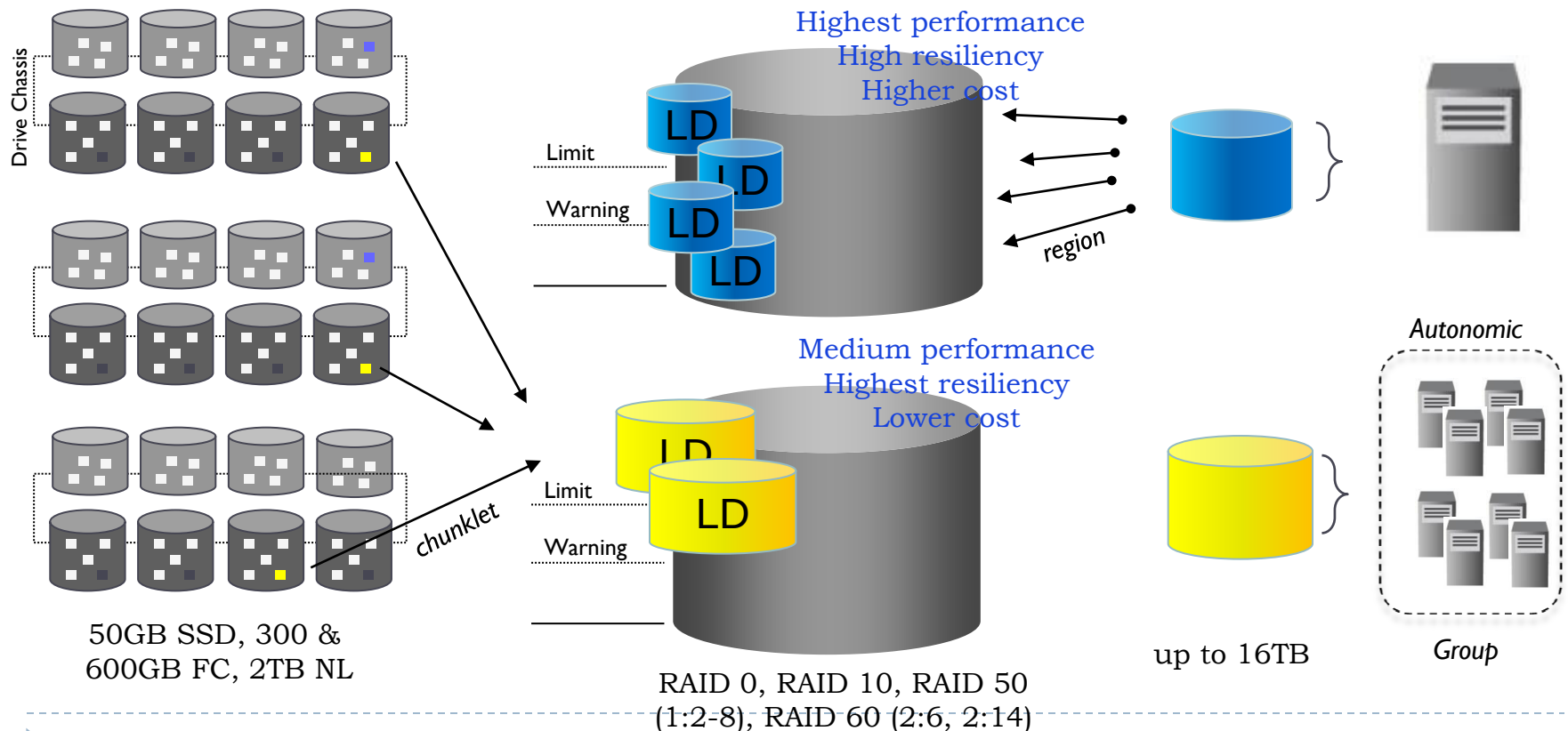
Physical drives broken into fine-grained uniform chunklets

Common Provisioning Groups

Automatically grows by creating Logical Disks (LDs) as needed (parameters set by the CPG policies)

Virtual Volumes

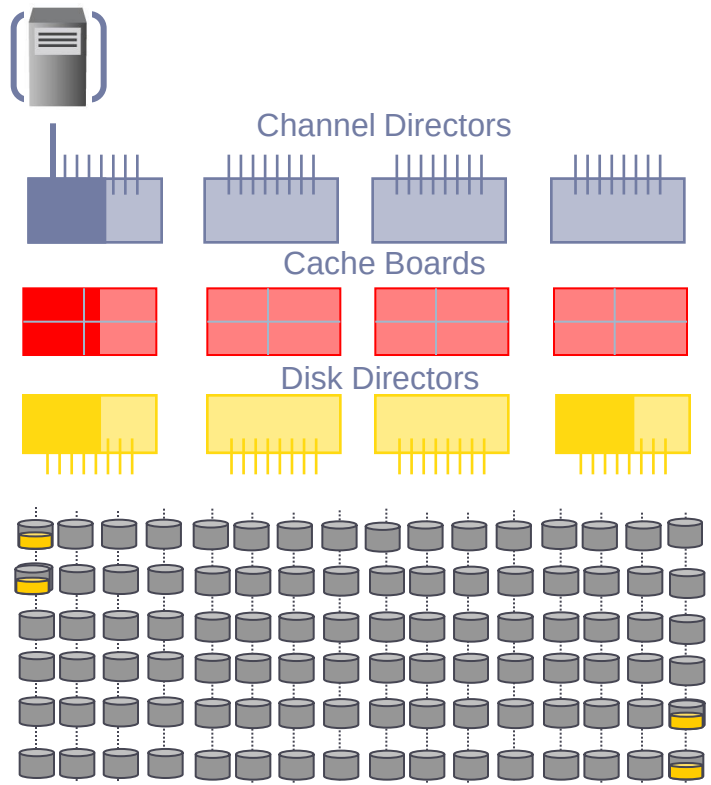
Created and exported in two commands—15 seconds with no pre-planning



3PAR Wide Striping

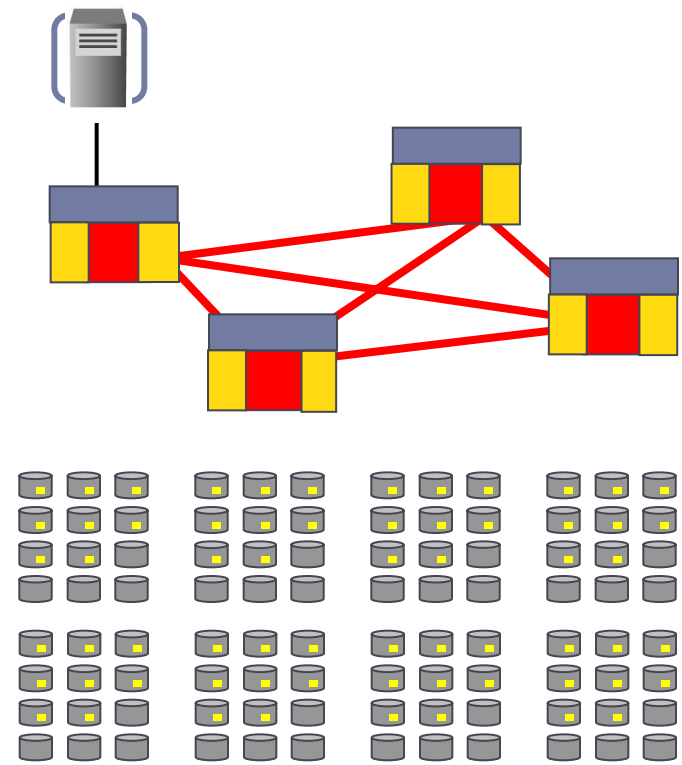
Completely and finely load balanced

Market Leading Monolithic Array



8GB LUN spread across 4 disks (RAID 10)

StorServ Storage Server



8GB LUN spread across 64 disks (RAID 10)

Why are Chunklets so Important?

- ▶ Same drive spindle can service many different LUNS, and many different RAID types at the same time
- ▶ Enables array to be managed by policy, not by administrative planning
- ▶ Drives up utilization rate of the disks (no leftover space)
- ▶ Improved high availability and sparing
- ▶ Enables easy mobility among physical disks, RAID types, service levels



High Availability + High Utilization Rate

HP 3PAR availability level concepts

▶ HA: Cage

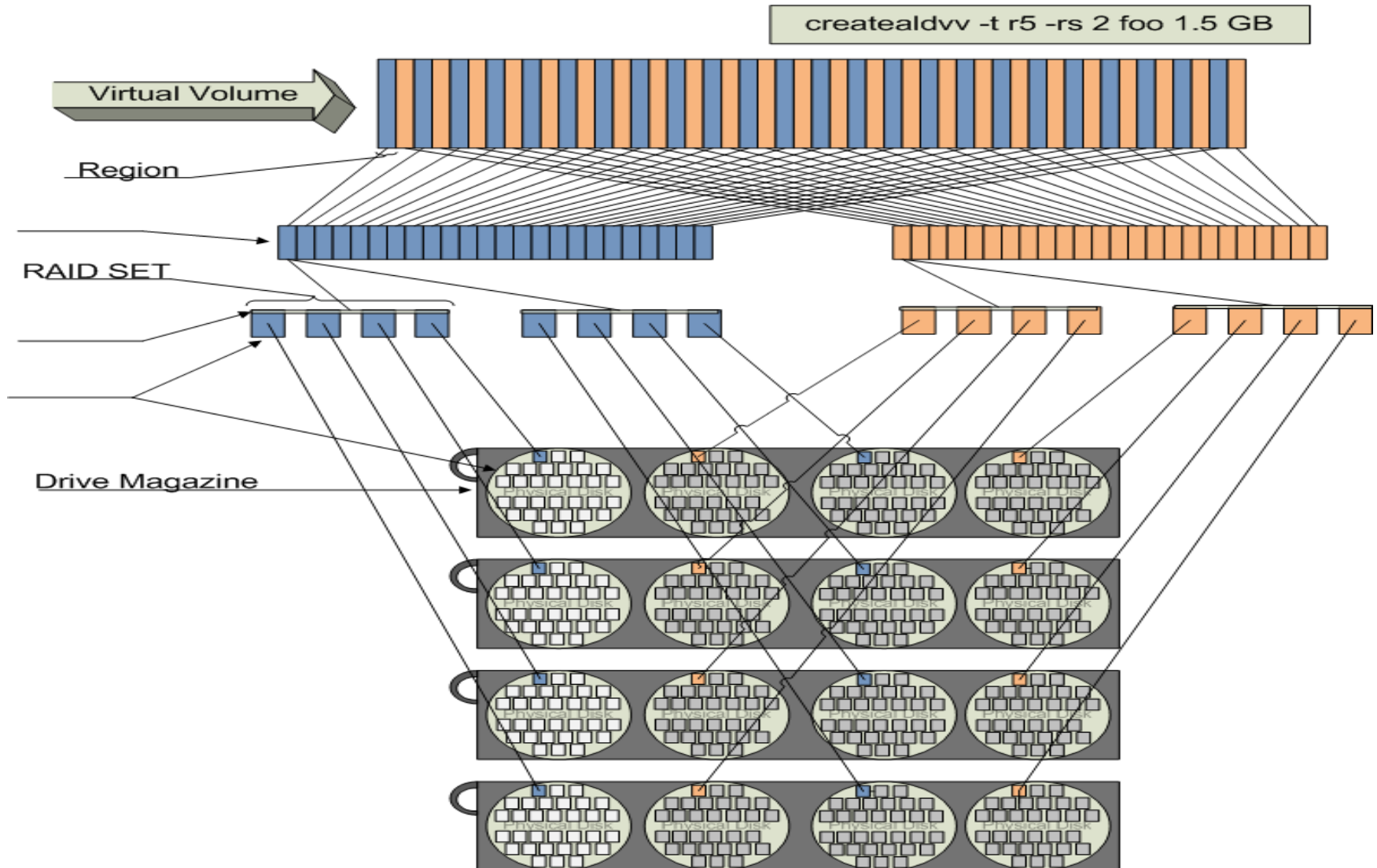
▶ HA: Magazine

▶ HA: Port

- These are policies by which the system lays out RAID sets in order to protect against hardware failures
- The default configuration will always be the “safest” but can be overridden by an administrator
- Default HA level “cage” unmatched in industry



How a Volume Maps from VV to Chunklet



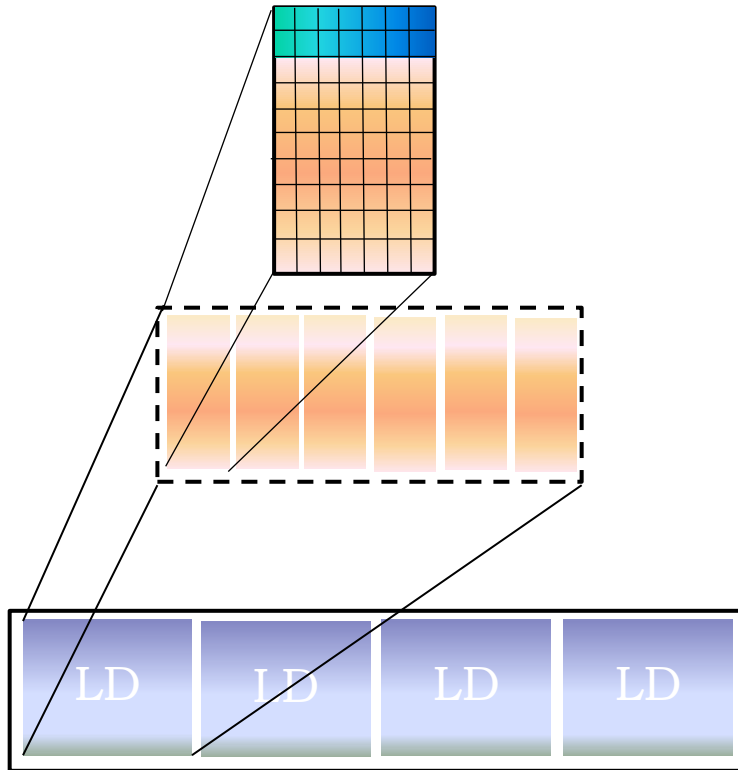
The 3 levels of capacity allocation

Pages
16KB

Regions
128MB

CPG
32GB *

Chunklets
256MB/1GB



Regions are divided into pages

LDs are divided into regions. Regions are allocated to only one VV

Chunklets are assembled in LDs to form the CPG

Multiple VVs will share the CPG

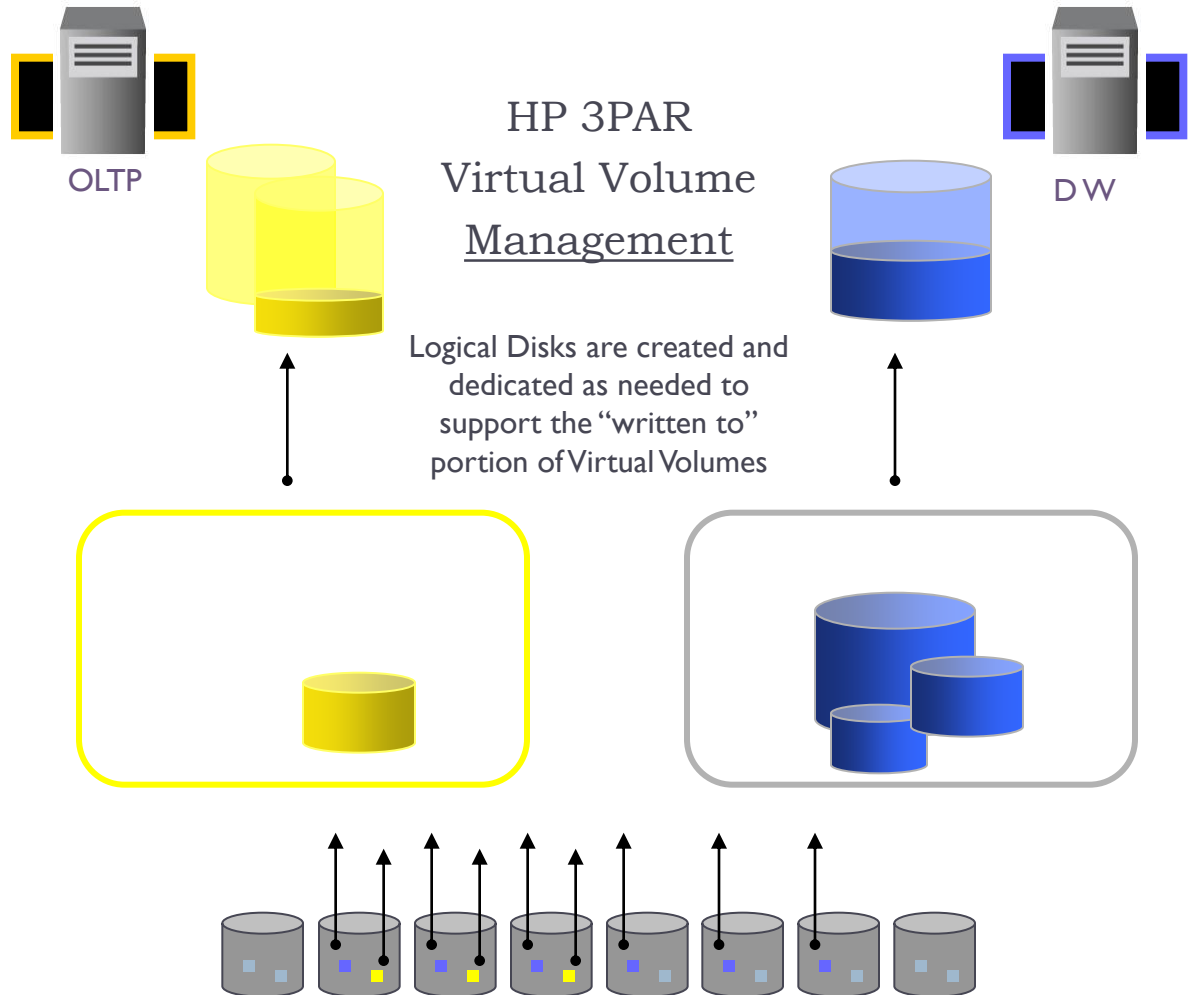
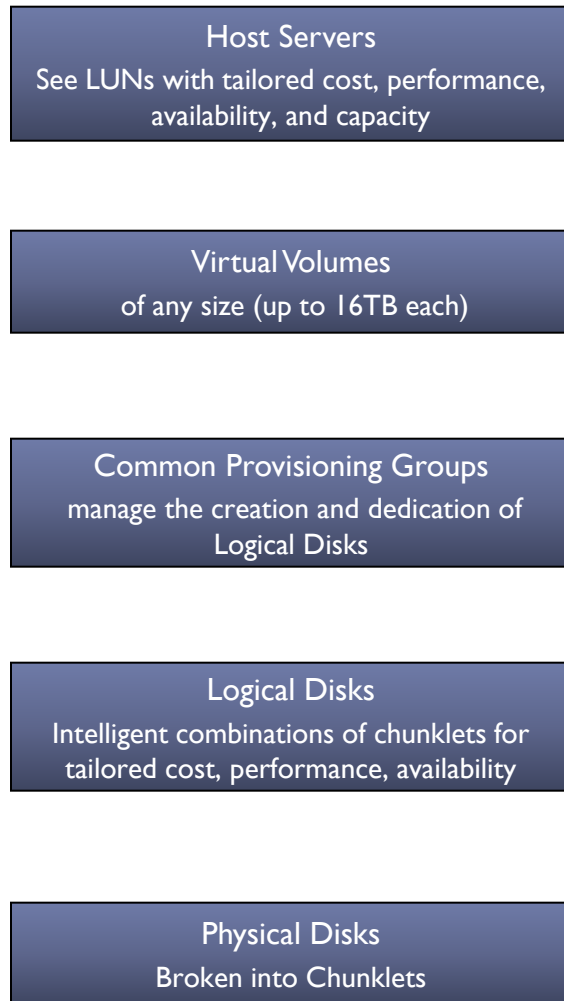
* default

Common Provisioning Groups (CPG)

- ▶ CPGs are FUNDAMENTAL to administering HP 3PAR
- ▶ CPGs automatically provision Logical Disk capacity on demand
- ▶ CPGs are the combination of a RAID type and a drive type which equals SERVICE LEVEL and AVAILABILITY LEVEL
- ▶ CPGs have many functions
- ▶ They are:
 - The policies by which free chunklets are assembled into logical disks
 - A container for existing volumes and used for reporting
 - The basis for service levels and our optimization products



Common Provisioning Groups (CPG)



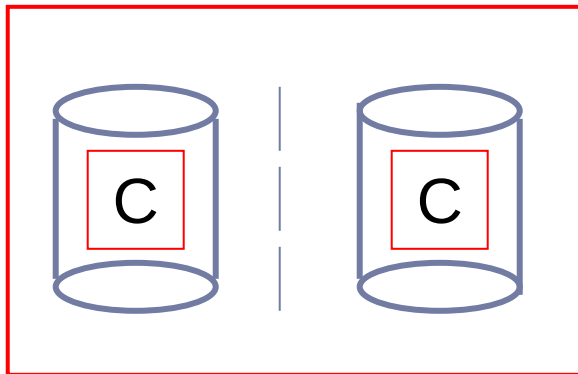
3PAR RAID Levels

- ▶ 3par OS can assemble RAID sets (Raidlets) into RAID 00, RAID 10, RAID 50, RAID 60
- ▶ Data to Parity Ratios (3PAR Setsize)
- ▶ Raid 10 mirror or 3-way mirror
- ▶ Raid 50 from 2+1 to 8+1 (HA-magazine)
- ▶ Raid 60 at 6+2 or 14+2 (protect against double disk failure with same capacity tradeoff of RAID 50)



3PAR RAID 10 Concepts

- ▶ RAID 10 is “mirrored” data
- ▶ Data is written as “paired” chunklets
- ▶ Each “chunklet” on the RAID set is on a different physical disk



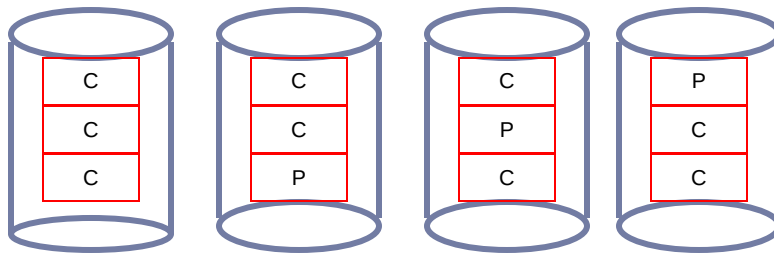
Setsize = 2

Default size (RAID 1)

Usable space_
= 1*chunklet size
(256MB or 1GB)

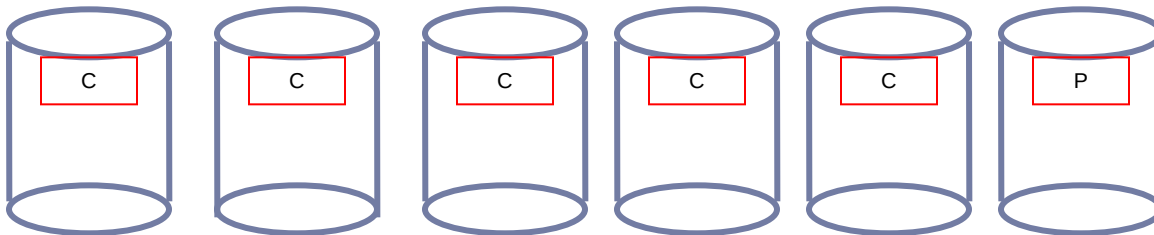
3PAR RAID 5 Concepts (1 of 3)

- ▶ RAID 5 uses parity to reconstruct data
- ▶ RAID 5 uses a setsize equal to number of drive cages (shelves) by default to guarantee cage level HA



Setsize = 4 (3+1)
Default size (RAID 5)

Usable space =
3*chunklet size
(768MB or 3GB)

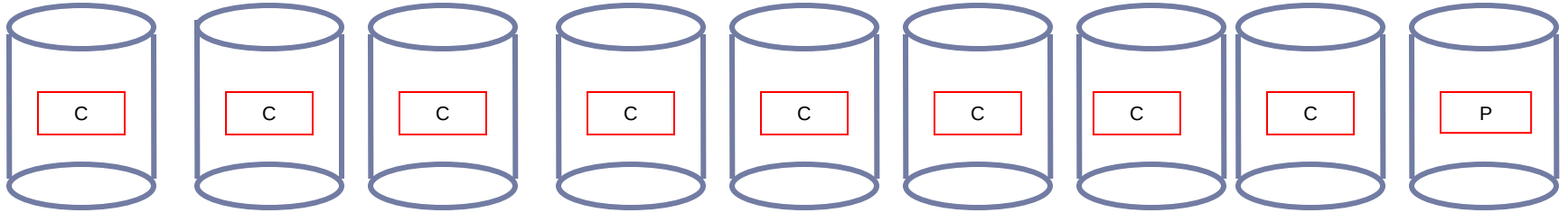


Setsize = 6 (5+1)

Usable space =
5*chunklet size
(1280MB or 5GB)



3PAR RAID 5 Concepts (2 of 3)

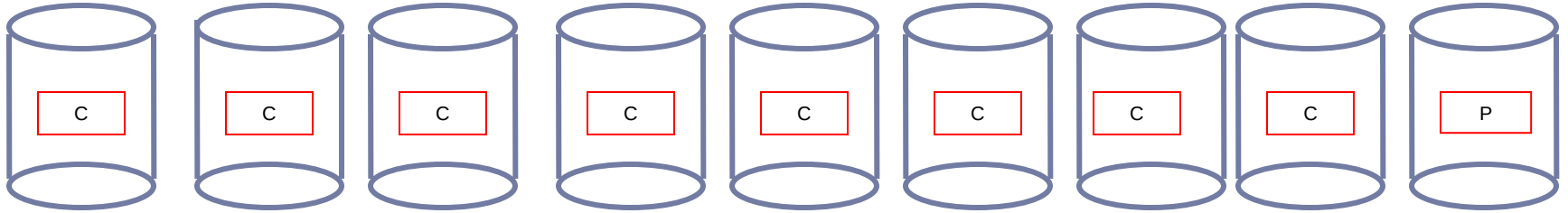


Setsize = 9 (8+1)

What is the usable space for this setsize?



3PAR RAID 5 Concepts (3 of 3)



Setsize = 9 (8+1)

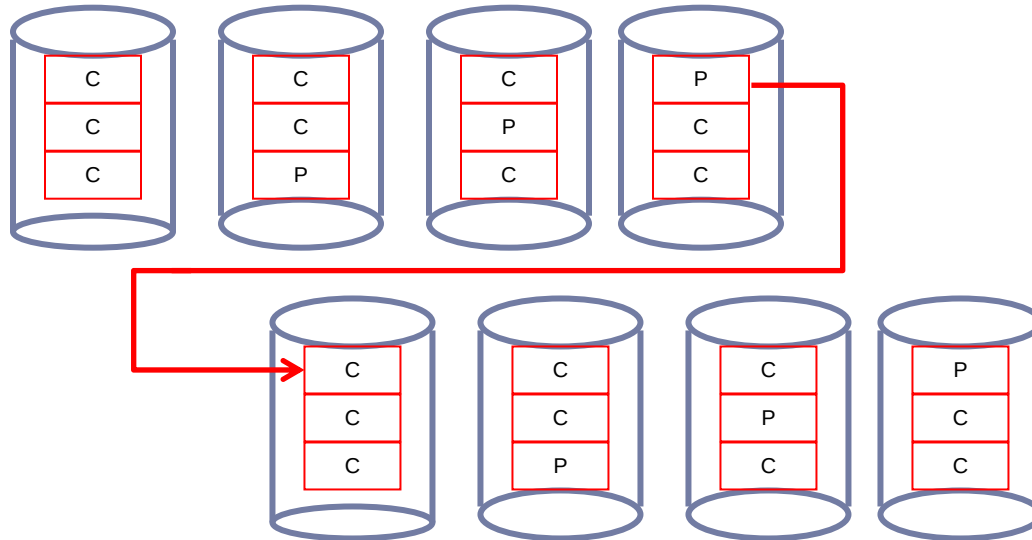
What is the usable space for this setsize?

Usable space = 8*chunklet size (2048MB or 8GB)



3PAR RAID Multi-Parity (MP) Concepts

- ▶ RAID MP uses parity (double parity—can deliver data in a double disk failure) to reconstruct data and perform in the ASIC XOR engine
- ▶ RAID MP supports only two set sizes (8 and 16)
- ▶ The default set size of 8 has the same data-to-parity ratio as the default RAID-5 set size of 4 -- 3:1 in both cases



Setsize = 8 (6+2)
Default size (RAID 6)
Usable space =
6*chunklet size
(1536MB or 6GB)

Usage—Meaning of “-ha cage” For RAID MP

- ▶ The system defines “-ha cage” as “will tolerate the failure of ONE cage.”
- ▶ This means:
 - Up to two chunklets are allowed to share the same cage
 - The default R6 “-ha cage” setsize 8 requires 4 cages per node-pair, just like the default R5 setsize 4
 - If 8 cages are available, the layout will use one chunklet per cage
 - The same rules apply to “-ha mag”—up to two chunklets are allowed per magazine, but the system will place only one chunklet per magazine if possible



Virtual volume types

- ▶ There are several types of virtual volumes:
 - ▶ • Fully Provisioned Virtual Volume (FPVV)
 - ▶ • Thinly Provisioned Virtual Volume (TPVV)
 - ▶ • Administrative Volume (created by the system and for system usage only)



Virtual Volumes

- ▶ The provisioning type classifies a VV as one of the following:
- ▶ **FPVV**—Fully provisioned VV with space for the base volume allocated from the associated user CPG and snapshot space allocated from the associated snapshot CPG.
- ▶ **TPVV**—Thinly provisioned VV with space for the base volume allocated from the associated user CPG and snapshot space allocated from the associated snapshot CPG (if any).
- ▶ **TDVV**—Thinly deduplicated VV with space for the base volume allocated from the associated user CPG and snapshot space allocated from the associated snapshot CPG (if any).
- ▶ **Administrative volumes**—Created by the system for system use only.



Data Components

- ▶ FPVVs, TPVVs, and TDVVs have three separate data components:
- ▶ **User space** is the area of the volume that corresponds to the LD regions in the CPG that are available to the host. The user space contains the user data and is exported as a LUN to the host.
- ▶ **Snapshot space** (or copy space) is the area of the volume that corresponds to LD regions in the CPG that contain copies of user data that changed since the previous snapshot. The snapshot space contains the copy data.
- ▶ **Administration space** (or admin space) is the area of the volume that corresponds to LD regions in the CPG that track changes to the volume since the previous snapshot was created. The administration space contains pointers to copies of user data in the snapshot space.



Dual –Snapshot Data (SD) for TPVVs

- ▶ TPVVs are comprised of two unique snapshot data (SD) spaces – one for TPVV written data and the other for modified data associated with snapshots.
- ▶ TPVVs can still be associated to one CPG
- ▶ SD spaces can be configured in either a single CPG or two separate CPGs for variable Quality of Service (QoS).



Dual SD Space

- ▶ When creating a TPVV, the user may associate the base virtual volume SD space with one CPG and virtual copy SD space with a separate CPG.
- ▶ When creating a base virtual volume with the “createvv” command the “-snp_cpg” option is used to specify the virtual copy SD space.

▶ For example:

```
createvv -tpvv -snp_cpg cpg1 cpg2 mytpvv 10g  
showvv -cpgalloc mytpvv
```

Id	Name	Prov	Type	UsrCPG	SnpcPG
28	mytpvv	tpvv	base	cpg2	cpg1



3PAR Virtual Volume

A virtual disk. Assembled by policy. Backed by every physical disk of the specified service level.



Host sees Virtual Volume as
a LUN

The only storage component visible to Hosts

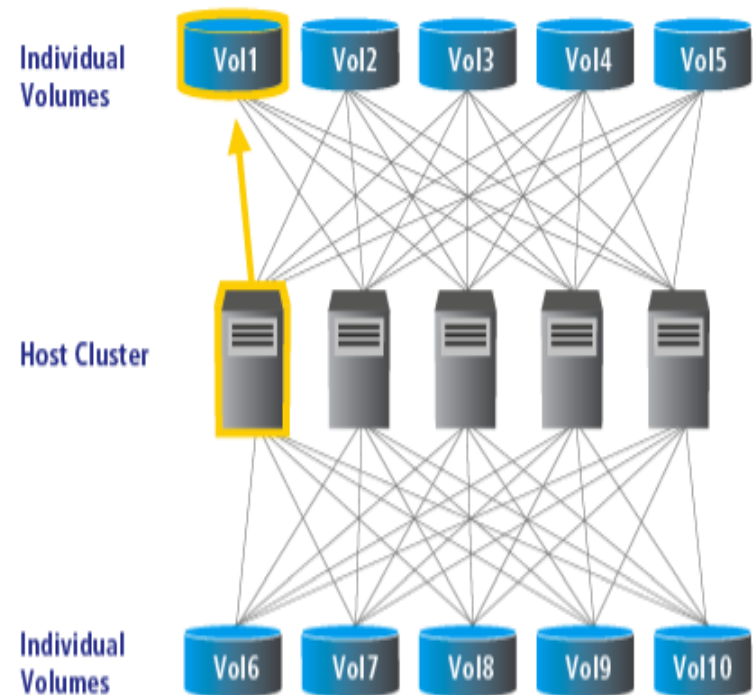




Clustered environments add additional complexity to storage provisioning

- ▶ VMware clusters create new storage management challenges
- ▶ 1 cluster of 5 hosts and 10 volumes, requires 50 provisioning actions on most traditional arrays!
 - At 10 min/action: More than 1 day!
 - Error-prone
- ▶ VMware clusters are dynamic resources subject to growth and frequent change

TRADITIONAL ARRAYS





3PAR Autonomic Groups eliminate repetitive error-prone tasks

- ▶ **3PAR Autonomic Groups**
 - Simplify and automate volume provisioning in a single command
- ▶ **Exports a group of volumes to a host or cluster of hosts**
 - Automatically preserve same LUN ID for each volume across hosts
- ▶ **When a new host is added into the host group:**
 - All volumes are autonomically exported to the new host
- ▶ **When a new volume is added into the volume group:**
 - New volume is autonomically exported to all hosts in the host group

3PAR AUTOMONIC PROVISIONING

Autonomic Host Group



Autonomic Volume Group



With 3PAR Autonomic Groups: 3 clicks and 60 seconds to fully create and provision multiple volumes to multiple servers.



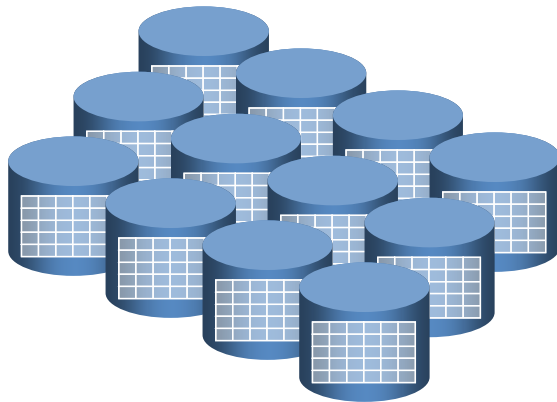
Host Groups

- ▶ Virtual Volumes (VVs) can be exported to multiple hosts
 - As a member of an exported VV Autonomic group and/or
 - As VV exports to each individual host
- ▶ Removing the Group export does not affect any exports to individual hosts
- ▶ Visible in showvln etc with the “set:” prefix in the hostname for the template and “host set” as the export type
 - The active export shows the individual host with a “host set” export type

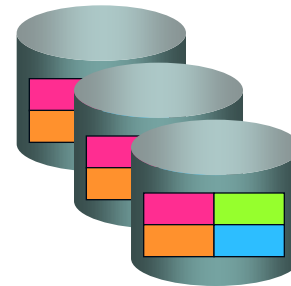


Replication of Database Applications

- ▶ A database application may be spread out over numerous files, file systems, and devices, all of which must be replicated
- ▶ Database replication can be offline or online



Data



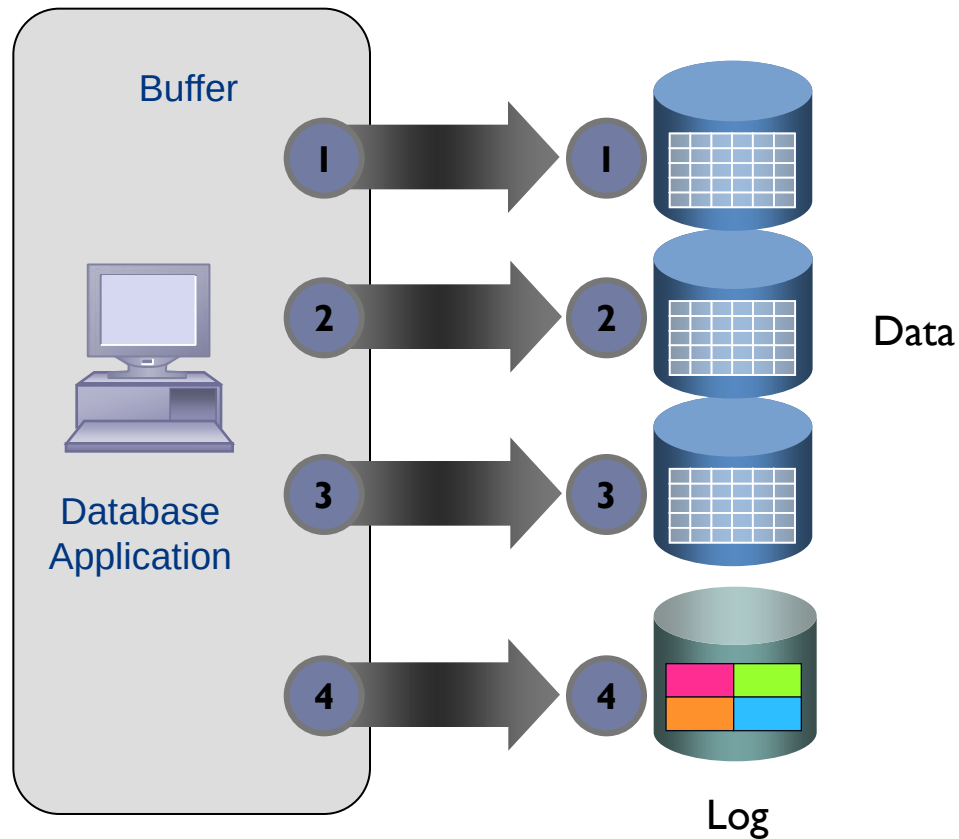
Logs

Database: Understanding Consistency

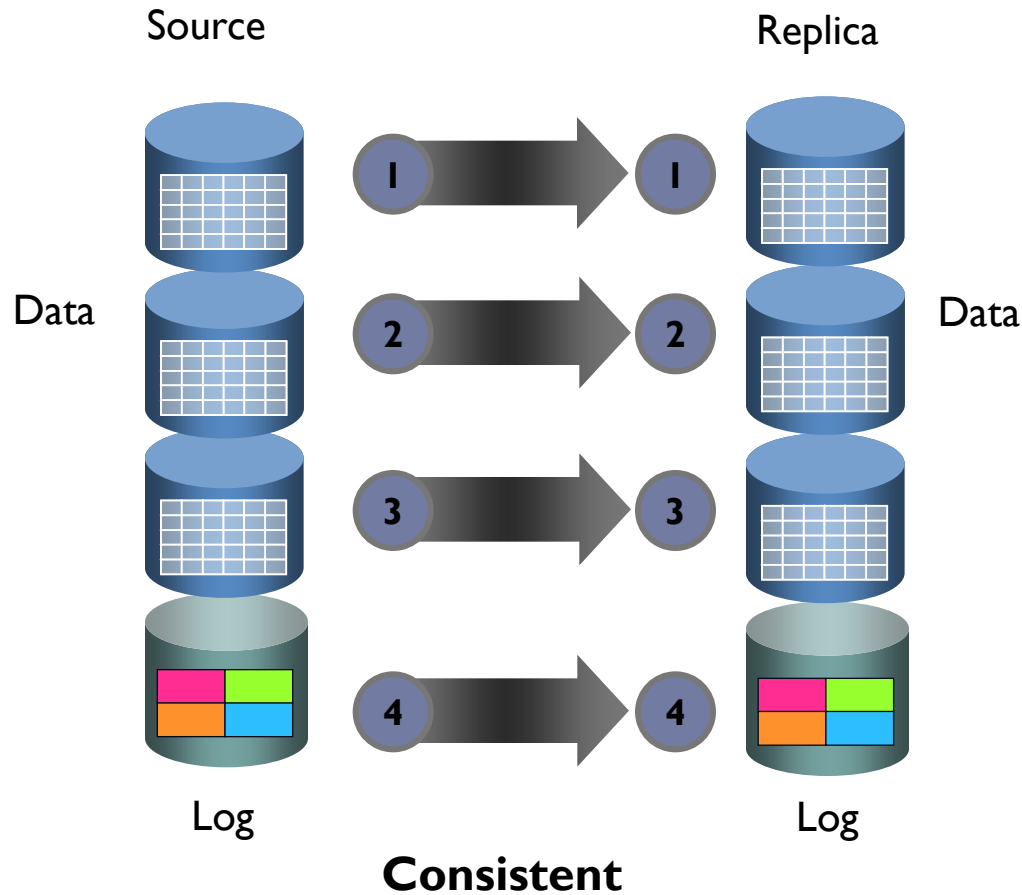
- ▶ Databases/Applications maintain integrity by following the “Dependent Write I/O Principle”
 - ▶ Dependent Write: A write I/O that will not be issued by an application until a prior related write I/O has completed
 - ▶ A logical dependency, not a time dependency
 - ▶ Inherent in all Database Management Systems (DBMS)
 - ▶ e.g. Page (data) write is dependent write I/O based on a successful log write
 - ▶ Applications can also use this technology
 - ▶ Necessary for protection against local outages
 - ▶ Power failures create a dependent write consistent image
 - ▶ A Restart transforms the dependent write consistent to transactionally consistent
 - i.e. Committed transactions will be recovered, in-flight transactions will be discarded



Database Replication: Transactions

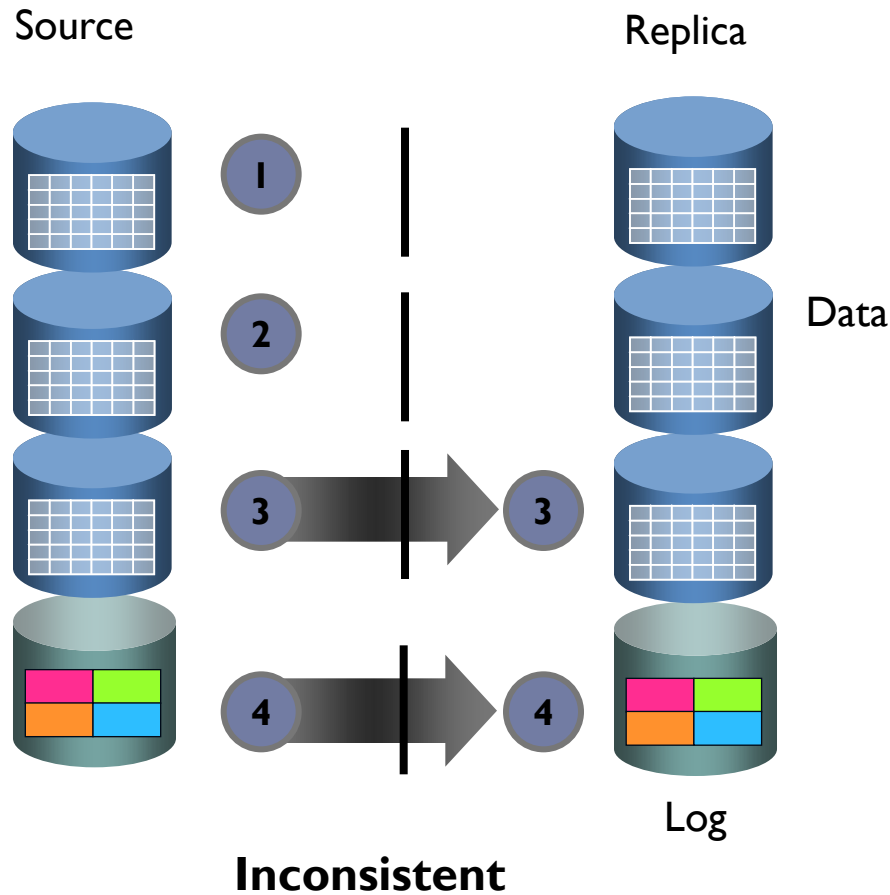


Database Replication: Consistency



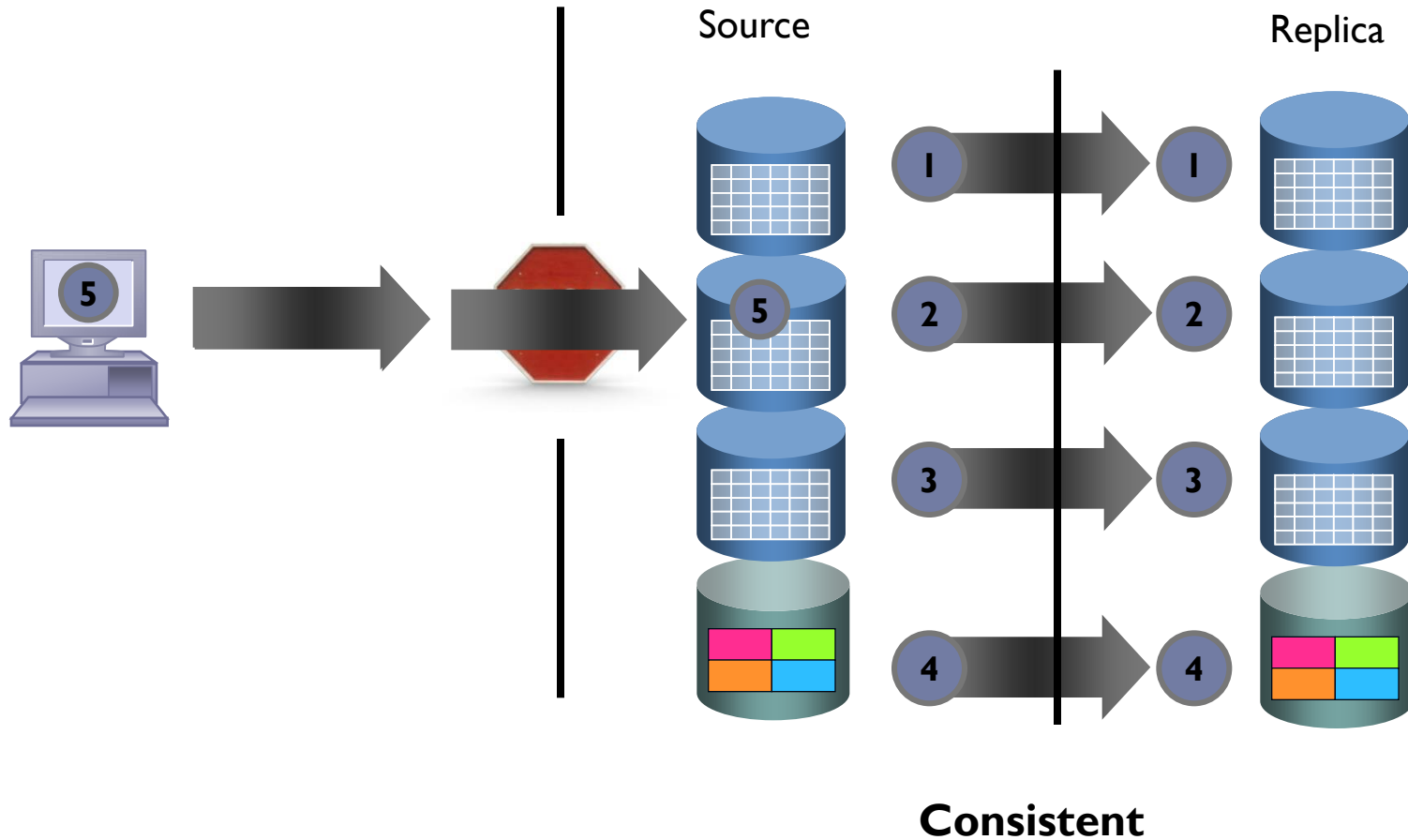
Note: In this example, the database is online.

Database Replication: Consistency



Note: In this example, the database is online.

Database Replication: Ensuring Consistency



Thank You ...

